Presentation 2

What did each of the team members do?

Henry: Mandatory programming tasks 1-4, 8 and task 11 (patch image filename standarization, inspect data quality, inspect distribution of slices, create new cases from cases with too many slices in a stain)

Vicky: two moderate tasks 5 and 6, add entropy regularization for attention scores to avoid peaky attention scores, and use gated attention instead of the current one.

How does it help the project?

Henry: It creates a more robust data input that identifies unusual files and data exploration in python notebook finds the distribution of every slice in a stain.

Vicky: The newer model increases test accuracy to 0.7895, decreases test loss to 0.5891.

Gated Attention improves accuracy by introducing nonlinearity, allowing the model to capture complex diagnostic patterns that simple linear attention cannot detect.

Entropy Regularization improves stability and lowers loss by preventing the model from concentrating too heavily on a few noisy patches, encouraging a more balanced evaluation of the entire tissue slide.

Issues faced that haven't been resolved:

- Current pseudocase function works but the code is messy and could be improved to avoid data leakage
- Training currently runs on CPUs, taking ~15 minutes per epoch. This limits extensive hyperparameter tuning (e.g., broader λ search) and efficient 5-fold cross-validation.

Proposed next steps:

- New results (was not ready in time for recording) look very good but could be deceptive. Need to investigate the splits. Also create a better solution to view what the number of slices are for each case after data augmentation and stratified splitting.
- Use GPU instead of CPU to train models.

References

KimiaNet Github↗

Links to the presentation & code:

- **Presentation:** **Presentation2_Team6**
- **Code:** **Code2_KimiaNet_10Feb**